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SUPPLEMENTARY INFORMATION
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Drought alters the structure and functioning of complex food webs

Supplementary Method 1

Stream mesocosms. Each mesocosm was a linear channel (width 0.33 m, length 12 m, depth 0.30 m) receiving water and suspended particles (including algae, detritus, and invertebrates) through a 110 mm diameter feeder pipe (6 m length). Water flow was controlled by a valve at the closed upper end of each channel. Water drained freely from mesocosms under gravity, via an open outlet positioned 10 cm above a downstream channel to prevent any potential cross-contamination among the mesocosms. Physicochemistry was highly congruent among mesocosms and closely paralleled those of the source stream[31]. Biota (algae and macroinvertebrates) in the mesocosms were taxonomically diverse and similar in composition to nearby streams[32]. Channels were filled with a 20 cm layer of stony substrate of the same substratum particle size distribution (85 % of particle volume 11-25 mm) and geological parent material (chert) to that of the source stream[19,21], providing both benthic and interstitial substrata in which suitably-adapted species may find refuge during drought[31-32]. The depth of mesocosm sediments was consistent with that of oxygenated hyporheic sediments (<20 cm) in neighbouring streams[33] and within the range of published estimates of the depth of hyporheic zones[34-36], but our mesocosms may be less representative of systems with deeper hyporheic zones and taxa that can avoid desiccation by migrating in to these deep sediments. There is however no general consensus as to the importance of the hyporheic zone as a refugium for biota during stream disturbances[37], including severe hydrologic droughts[38]. For example, a recent review[38] revealed that of the 18 studies testing unequivocally for hyporheic drought refugia, half found no evidence of refuge use, while the other half found support for the hyporheic refuge hypothesis. Thus, our results should be applicable to a wide range of systems, including headwaters with shallow sediments and lowland streams where a range of natural (e.g. geomorphology, colmation) and anthropogenic (e.g. land-use practices, contamination, hydro-engineering) factors can limit the oxic depth of the hyporheic zone.

Supplementary Method 2

Macroinvertebrates: sample processing and secondary production estimation. Animals in samples were sorted from debris, identified to the lowest practicable taxonomic unit (usually species or genus) and counted. For secondary production estimation, macroinvertebrate body lengths (all sampled individuals, n=63,092) were measured to the nearest 0.1 mm using an ocular graticule and dissecting microscope. Individual biomass (g dry weight) was calculated for all macroinvertebrate specimens using published length-mass regressions[39]. Secondary production of all macroinvertebrates was calculated from biomass using the size-frequency method[28] (excepting rare taxa < 1 % total abundance where production was estimated by multiplying mean annual biomass by an annual P/B value of the most closely related taxon[19]). Production for the first year and the second year of the experiment was averaged and incorporated in to biomass flux estimates as mean annual secondary production (g m^{-2} yr^{-1}).

Supplementary Methods 3

Food web construction. Binary food webs were constructed based on the presence/absence of resources in the diet of consumers sampled at the end of the experiment. These webs were then quantified, with links expressed as flows of biomass from resources to consumers for each mesocosm community. The trophic basis of production method[19] was used to quantify directly observed feeding links, with biomass flux (F_{ij}, g m^{-2} yr^{-1}) from resource *i* to consumer *j* estimated as follows:

Determine the proportion of production derived from food type *i* (B_i):

$B_i = (G_i \times AE_i) / \Sigma_{Gi=1,\ldots,n}$.

Calculate the flow of biomass via food type i to consumer j (F_{ij}).

$F_{ij} = (B_i \times P_j) / (AE_i \times NPE)$

where G_i is the percentage cover of food type i, AE_i is the assimilation efficiency of food type i, P_j is the secondary production[19] of consumer j, and NPE is assumed net production efficiency.

Feeding linkages were determined directly by gut contents analysis (x 1000) of macroinvertebrates. The food webs were dominated by herbivore-detritivores that feed on ubiquitous detritus and microalgae, and our instantaneous sample of diet was characteristic of feeding throughout the year[40]. In total 3,643 dissected guts were examined, with consumed items identified to the lowest practicable taxonomic unit. The guts of invertebrates were dissected at x20 magnification, and the gut contents were mounted on glass slides with an aqueous agent (Aquamount®). Five fields of view were examined on each slide at x 200 magnification using an ocular grid (1 cm^2 divided into 100 cells of 1 mm^2). Gut contents were identified as algae, fungi, invertebrates, large plant detritus and amorphous detritus. Amorphous detritus is organic matter derived from biofilms on the stream bed. It consists of polysaccharide matrix, microorganisms and their by-products[18]. Invertebrate, diatom and other algal components of diet were identified to genus or species whenever possible. The relative amount of each food type in a field of view was derived by counting the squares on the ocular grid dominated by that food type. The percentage of each food type for an individual was then calculated from the five fields of view and expressed as a percentage of the total particle area. Yield effort curves (number of food types versus number of guts examined) were drawn for each taxon to determine when a sufficient number of individuals had been examined to describe its diet accurately[27].

Supplementary Method 4

Quantitative food web metrics. Food webs with links quantified as flux of biomass (g m^{-2} yr^{-1}) from resources to consumers were compared using metrics derived from information theory[5,20]. For each food web, we determined the quantified, weighted measures of linkage density (*LDq*), interaction diversity (*IDq*), interaction evenness (*IEq*), generality (*Gq*, mean number of resources per comsumer) and vulnerability (*Vq*, mean number of consumers per resource). The metrics incorporate the inflow and outflow of biomass to each species in the food web, and the diversity of biomass flows derived from the resource (H_N, the diversity of inflows) and going to the consumers (H_P) of each taxon k was calculated as:

$$H_{N,k} = -\sum_{i=1}^{s} \frac{b_{ik}}{b_{\bullet k}} \log_2 \frac{b_{ik}}{b_{\bullet k}}$$

$$H_{P,k} = -\sum_{j=1}^{s} \frac{b_{kj}}{b_{\bullet k}} \log_2 \frac{b_{kj}}{b_{\bullet k}}$$

In each food web matrix, column sum $b_{\bullet k}$ and row sum $b_{k\bullet}$ are the sum total biomass flux from resources, and to consumers, of taxon k, respectively. The reciprocals of $H_{N\bullet k}$ and $H_{P\bullet k}$ are:

$$n_{N,k} = \begin{cases} 2^{H_{N,k}} \\ 0 \end{cases} \text{if } b_{\bullet k} = 0$$

$$n_{P,k} = \begin{cases} 2^{H_{P,k}} \\ 0 \end{cases} \text{if } b_{k\bullet} = 0$$

Weighted quantitative linkage density (LD_q) was calculated as the average of the equivalent numbers of resources ($n_{N,k}$) and consumers ($n_{P,k}$), weighted by their inflows and outflows:

$$LD_q = \frac{1}{2}\left(\sum_{k=1}^{s} \frac{b_{k\bullet}}{b_{\bullet\bullet}} n_{P\,k} + \sum_{k=1}^{s} \frac{b_{\bullet k}}{b_{\bullet\bullet}} n_{N\,k} \right)$$

where $b_{\bullet\bullet}$ is the total biomass flux in the web matrix[20]. Quantified connectance was calculated as LD_q/S. Weighted generality (G_q) and vulnerability (V_q) were calculated as:

$$G_q = \sum_{k=1}^{s} \frac{b_{\cdot k}}{b_{\cdot\cdot}} n_{N,k}$$

$$V_q = \sum_{k=1}^{s} \frac{b_{k\cdot}}{b_{\cdot\cdot}} n_{P,k}$$

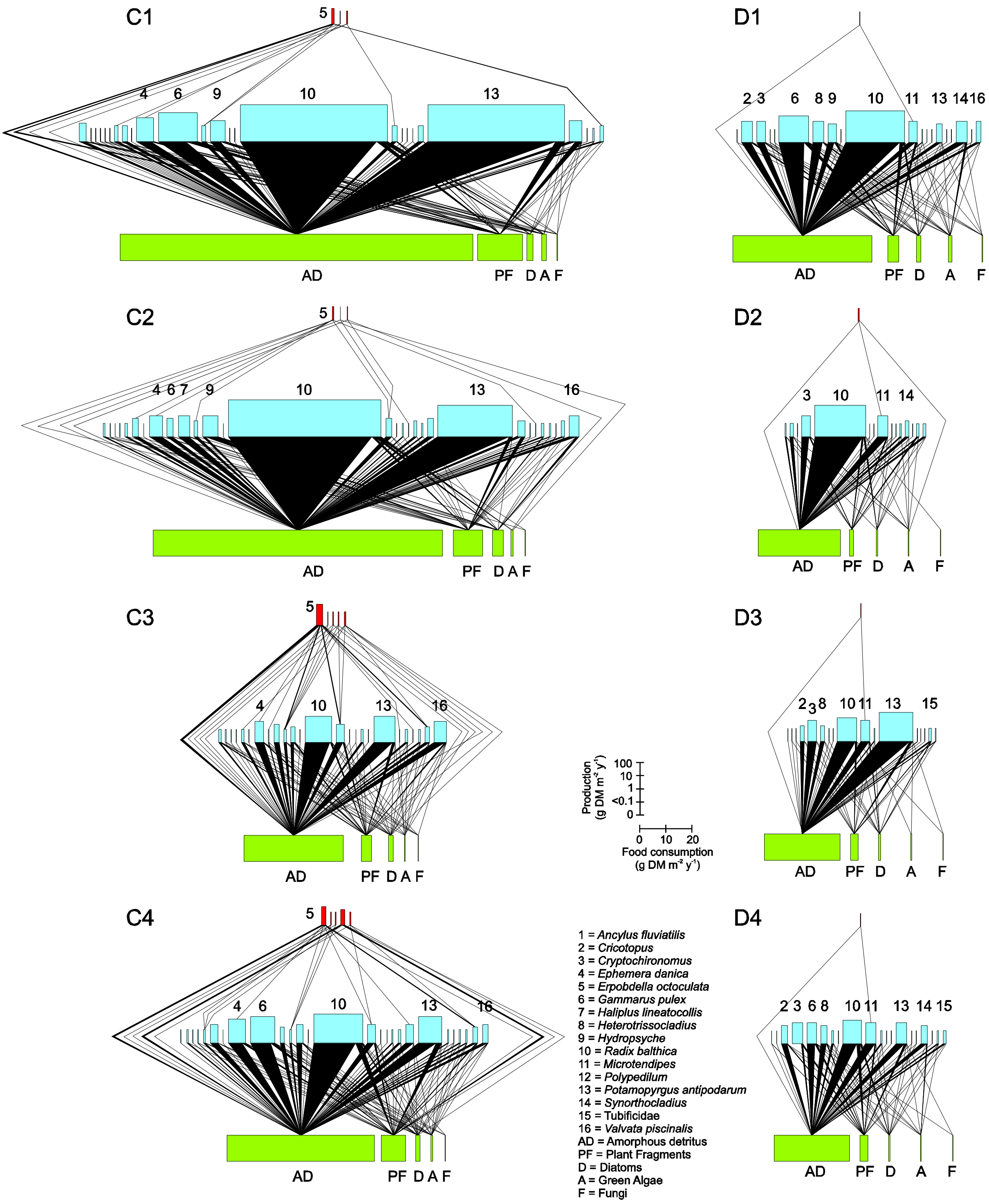
The diversity and evenness of quantified links in each food web was calculated using the Shannon index of entropy:

$$ID_q = \sum p_i \log_2(p_i)$$

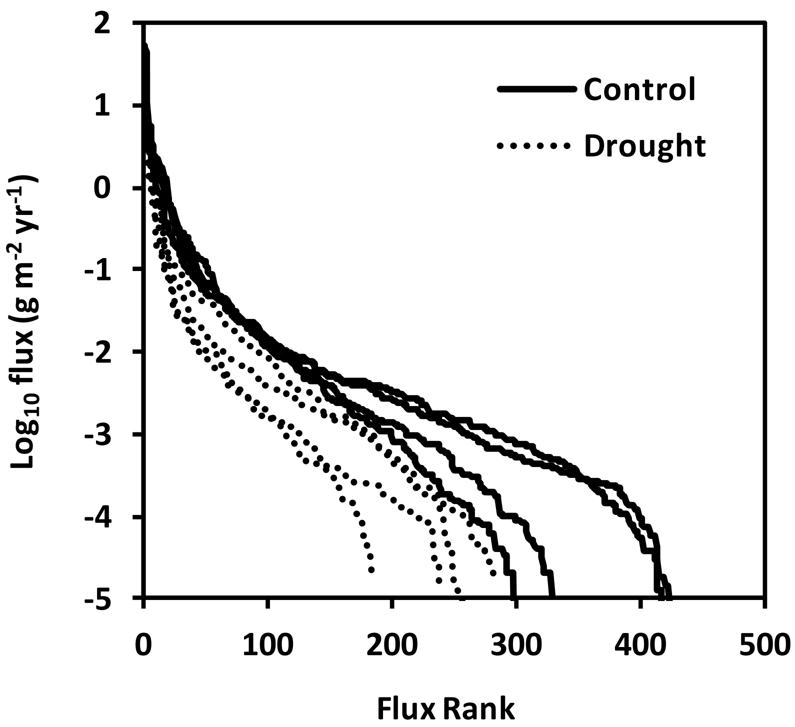
$$IE_q = \frac{\sum p_i \log_2(p_i)}{\log_2 N}$$

Where p_i is the proportional contribution of interaction i to the total number of interactions in the web (N).

Supplementary Figure 1. Quantitative food webs in replicate control (C1-C4) and drought (D1-D4) treatments (webs C1 and D1 shown in Fig. 1). For each web, lower bars are basal resources, middle bars are primary consumers and top bars are predators. For each consumer, the height and width of the bars is proportional to mean annual secondary production and biomass flux from resources (total inflows), respectively. For basal species, the relative width of bars on the x-axis is proportional to total consumption by invertebrates (total outflows from each resource to consumers), and for this trophic level production (y-axis) was not quantified. The black triangles that link trophic levels illustrate the relative contribution of resource flows to the production of each consumer, summing to the total inflows. Numbers refer to consumer identity and letters distinguish categories of basal resource, omitting rare species (<1% total production). Flows from individual algal taxa are grouped for display only. See Supplementary Tables 1 and 2 for full lists of resource and consumer taxa, respectively.



Supplementary Figure 2. Magnitude of biomass fluxes from resources to consumers for all control and drought-disturbed webs. For each treatment, fluxes were ranked from left to right in order of decreasing magnitude.



Supplementary Table 1. List of benthic algae in mesocosm communities.

Group	Taxon
Bacillariophyceae	*Amphora inariensis* Krammer
	Amphora libyca Ehrenberg
	Amphora ovalis (Kützing) Kützing
	Amphora pediculus (Kützing) Grunow in Schmidt
	Cocconeis placentula Ehrenberg
	Cymatopleura solea (Brébisson & Godey) W. Smith
	Diatoma vulgare Bory
	Encyonema minutum (Hilse in Rabenhorst) Mann
	Fragilaria vaucheriae (Kützing) Petersen
	Gomphonema olivaceum (Hornemann) Brébisson
	Gyrosigma sp.
	Melosira varians Agardh
	Navicula capitata Ehrenberg
	Navicula gregaria Donkin
	Navicula lanceolata (Agardh) Ehrenberg
	Navicula menisculus Schumann
	Navicula tripunctata (O.F. Müller) Bory
	Nitzschia dissipata (Kützing) Grunow
	Nitzschia perminuta (Grunow) M. Peragallo
	Nitzschia sp. 1
	Placoneis clementis (Grunow) E.J. Cox
	Planothidium lanceolatum (Bréb. ex Kützing) Round & Bukhtiyarova
	Psammothidium lauenburgianum (Hustedt) Bukhtiyarova & Round
	Rhoicosphenia abbreviata (Agardh) Lange-Bertalot
	Staurosira elliptica (Schumann) Williams & Round
	Staurosirella leptostauron (Ehrenberg) Williams & Round
	Surirella brebissonii Krammer & Lange-Bertalot
	Surirella minuta Brébisson in Kützing
	Synedra ulna (Nitzsch) Ehrenberg
Chlorophyceae	*Gongrosira incrustans* Reinsch
Cyanophyceae	*Phormidium* sp. 1
	Phormidium sp. 2

Supplementary Table 2. List of macroinvertebrate taxa found in mesocosm food webs. Predatory taxa are highlighted in bold.

Group	Taxon
Oligochaeta	Naididae
	Tubificidae
Gastropoda	*Ancylus fluviatilis* (Müller)
	Potamopyrgus antipodarum (J.E. Gray)
	Radix balthica (L.)
	Theodoxus fluviatilis (L.)
	Valvata piscinalis (Müller)
Bivalvia	*Pisidium* sp.
Hirudinea	***Erpobdella octoculata (L.)***
Isopoda	*Asellus aquaticus* (L.)
Amphipoda	*Gammarus pulex* (L.)
Ephemeroptera	Baetidae
	Ephemera danica Müller
Plecoptera	*Leuctra geniculata* Stephens
Coleoptera	*Brychius elevatus* (Panzer)
	Elmis aenea (Müller)
	Haliplus lineatocollis (Marsham)
	Limnius volckmari (Panzer)
	Oulimnius tuberculatus (Müller)
	Platambus maculatus (L.)
Megaloptera	***Sialis lutaria*** **(L.)**
Trichoptera	*Athripsodes* spp.
	Brachycentrus subnubilus Curtis
	Hydropsyche spp.
	Limnephilus lunatus Curtis
	Polycentropus flavomaculatus **(Pictet)**
	Sericostoma personatum (Spence)
	Tinodes waeneri (L.)
Diptera	*Cricotopus* sp.
	Cryptochironomus sp.
	Heterotrissocladius sp.
	Macropelopia **sp.**
	Microtendipes sp.
	Pentaneura **sp.**
	Polypedilum sp.
	Procladius **sp.**
	Prodiamesa olivacea (Meigen)
	Synorthocladius sp.
	Simuliidae
	Tipula montium Egger

Supplementary Table 3. Qualitative (binary) food web metrics for drought and control stream food webs. Metrics were linkage density (*L*/*S*) where *L* is number of consumer-resource links and *S* is the number of species in the web, directed connectance (L/S^2), generality ($L/S_{consumers}$), vulnerability ($L/S_{resources}$). ANOVA tested for the effect of drought (below) and block (P >0.05, not shown).

	Control		Drought		ANOVA	
Metric	Mean	SE	Mean	SE	$F_{1,3}$	*P*
Linkage density	5.96	0.53	4.94	0.38	2.20	0.212
Directed connectance	0.09	0.01	0.10	0.01	0.17	0.706
Generality	11.68	1.11	12.84	0.55	1.42	0.390
Vulnerability	6.63	0.63	5.26	0.40	3.37	0.164

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